

Dr. Yonatan Ashenafi

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Education

Rensselaer Polytechnic Institute (RPI), Troy, NY

Ph.D. in Mathematics, May 2021

Thesis: *Stochastic Hydrodynamics of Colonial Microswimmers*

M.S. in Applied Mathematics, August 2016 – December 2018

Dordt University, Sioux Center, IA

B.A. in Mathematics, August 2012 – May 2016

Experience

Postdoctoral Fellow, Worcester Polytechnic Institute (WPI)

Aug 2023 -

Worcester, MA

- Conducting research on modeling intracellular transport processes using experimental data from multi-university collaborations.
- Teaching Multivariable Calculus, Calculus I & II, and Ordinary Differential Equations.

Postdoctoral Fellow, University of Alberta

Sep 2021 – May 2023

Edmonton, AB

- Researched motion mechanisms in diatoms (microscopic algae of ecological importance) through video-based data analysis.
- Taught Calculus I for Life Sciences and Calculus II for Physical Sciences.

Technical Fellow Intern, General Electric (GE) Research

Oct – Dec 2020

GE Global Research Center, Niskayuna, NY

- Collaborated with the Probabilistic Design Team on experimental design and additive manufacturing projects.
- Developed Python programs for reinforcement learning tasks.

Research Assistant, Dordt College

Jun – Aug 2014

Sioux Center, IA

- Constructed Gaussian mixture models from bacterial gene expression data using R.
- Gained experience in statistical data analysis and biological data modeling.

Tutor, Dordt College Library

Sep 2013 – May 2016

Sioux Center, IA

- Tutored students in College Algebra, Calculus I–III, Linear Algebra, and Discrete Mathematics.
- Collaborated effectively with students, faculty, and staff to improve academic performance.

Publications

- Ashenafi, Y. & Kramer, P. R. (2024). Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells. *Bulletin of Mathematical Biology*.
- Ashenafi, Y. (2022). Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin. *Physics of Fluids*.
- Ashenafi, Y., Pandita, P., & Ghosh, S. (2021). Reinforcement Learning-Based Sequential Batch-Sampling for Bayesian Optimal Experimental Design. *Journal of Mechanical Design*.
- Zheng, P. et al. (2023). Cooperative Motility, Force Generation and Mechanosensing in a Foraging Non-Photosynthetic Diatom. *Open Biology*, 13(10), 230148.
- Disselkoen, C., et al. (2016). A Bayesian Framework for the Classification of Microbial Gene Activity States. *Frontiers in Microbiology*, 7, 1191.
- Ashenafi, Y. & Kramer, P. R. Stochastic Analysis of Kinesis and Taxis for Colonial Protozoa. Under review.
- Ashenafi, Y. Linearizing Nonlinear Stochastic Dynamics via Bijective State-Space Transformations for Bayesian Tracking. In preparation.
- Ashenafi, Y. & Walcott, S. Simulation-Based Inference of Hidden Motor States in Intracellular Cargo Transport. In preparation.

Service and Activities

- Proposal Reviewer, National Science Foundation (NSF), 2025–present.
- Presenter, 2025 Canadian Mathematical Society Mathematics Education Meeting (Online).
- Judge, SIMIODE SCUDEM X 2025 (Differential Equations Modeling Challenge).
- Volunteer Grader, WPI Annual Math Meet, 2024.
- Presenter, Ethiopian Mathematics Professional Association Seminar, July 2025.
- Treasurer, Black Graduate Students Association, RPI (2020). Managed finances during the COVID-19 pandemic.
- Organizer, Mini-Symposium on Noise and Asymmetry in Microscopic Life, CAIMS 2022.
- Mathematical Modeler, Mathematical Problems in Industry (MPI) Workshop, Univ. of Vermont (2024) and RPI (2017).
- Participant, SIMIODE Spring 2024 Webinars.
- Presenter at Biophysical Society (2025),
- Presenter at Biology and Medicine through Mathematics (BAMM) conference (2024),
- Attended the Institute for Computational and Experimental Research in Mathematics (ICERM) Workshop on Interacting Particle Systems: Analysis, Control, Learning, and Computation (2024),
- Presenter at Society for Industrial and Applied Mathematics (SIAM) Life Sciences conference (2021),

- Presenter at Mathematical Biosciences Institute Workshop (2020),
- Presenter at Univ. of Minnesota Multiscale Modeling (2019),
- Judge, University of Alberta Undergraduate Research Symposium, 2022.

Awards

- Bill and Nancy Siegmund Applied Mathematical Modeling Prize, RPI (2021).
- DiPrima Summer Research Fellowship, RPI (2018).
- ASME CIE Advanced Modeling and Simulation Best Paper Award (2022).

Technical Skills

Python (NumPy, Pandas, SciPy, TensorFlow, scikit-learn), MATLAB, C++, R, SQL

Professional Membership

Society for Industrial and Applied Mathematics (SIAM)